Chapter 5. Logistic Regression

Interpretation, inference and fitting

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Logistic regression is the central model for binary responses.

5.1 Interpreting parameters in logistic regression

Motivation. People think in probabilities, so β needs several readings: odds ratio, local slope, change in π over x.

For binary Y and predictor X let $\pi(x) = P(Y = 1 \mid X = x)$. The logistic regression model is

$$\pi(x) = \frac{\exp(\alpha + \beta x)}{1 + \exp(\alpha + \beta x)}, \qquad \text{equivalently} \qquad \operatorname{logit} \pi(x) = \log \frac{\pi(x)}{1 - \pi(x)} = \alpha + \beta x. \tag{5.1}$$

5.1.1 Interpreting β: odds, probabilities, and linear approximations

The sign of β gives the direction; the curve steepens as $|\beta|$ grows and is flat at $\beta = 0$ (independence). For $\beta > 0$ it is a logistic cdf, approaching 1 and 0 at the same rate.

Proposition 5.1 (Odds-ratio interpretation)

Under (5.1) the odds are $\pi(x)/[1 - \pi(x)] = e^{\alpha}(e^{\beta})^{x}$; hence for every x,

$$\frac{\pi(x+1)/[1 - \pi(x+1)]}{\pi(x)/[1 - \pi(x)]} = e^{\beta} :$$

the odds multiply by e^{β} for each unit increase in x, and e^{β} is an odds ratio.

Proof. Exponentiate the logit form: odds $= \exp(\alpha + \beta x)$. The ratio at $x+1$ and x is $\exp(\alpha + \beta x + \beta)/\exp(\alpha + \beta x) = e^{\beta}$. ■

Proposition 5.2 (Linear approximation (Berkson 1951))

Under (5.1),

$$\frac{d\pi(x)}{dx} = \beta\, \pi(x)[1 - \pi(x)].$$

(a) The slope is greatest, $\beta/4$, where $\pi(x) = \frac{1}{2}$, i.e. at $x = -\alpha/\beta$ (the median effective level EL_{50}; in toxicology the LD_{50}). At $\pi(x) = 0.9$ or 0.1 it is 0.09β, and it tends to 0 as $\pi(x) \to 0$ or 1.
(b) $\pi(-\alpha/\beta \pm 1/\beta) = e^{\pm 1}/(1 + e^{\pm 1})$, i.e. 0.73 and 0.27: a change of $1/\beta$ in x from the EL_{50} changes π by about 0.23, close to the value $\frac{1}{4}$ suggested by the tangent line.

Proof. [Supplement]

Step 1. With $\eta = \alpha + \beta x$ and $\pi = e^{\eta}/(1 + e^{\eta})$,

$$\frac{d\pi}{d\eta} = \frac{e^{\eta}(1 + e^{\eta}) - e^{\eta} e^{\eta}}{(1 + e^{\eta})^2} = \frac{e^{\eta}}{(1 + e^{\eta})^2} = \pi(1 - \pi),$$

and $d\eta/dx = \beta$, so $d\pi/dx = \beta\pi(1 - \pi)$.

Step 2. The function $\pi(1 - \pi)$ on $(0, 1)$ is maximized at $\pi = \frac{1}{2}$ with value $\frac{1}{4}$; $\pi(x) = \frac{1}{2}$ iff $\alpha + \beta x = 0$ iff $x = -\alpha/\beta$. Also $0.9 \times 0.1 = 0.09$.

Step 3. At $x = -\alpha/\beta \pm 1/\beta$, $\eta = \pm 1$, so $\pi = e^{\pm 1}/(1 + e^{\pm 1}) = 0.731$ or 0.269. ■

So a unit change in x changes π by about $\beta\pi(1-\pi)$ (Figure 1). Alternatively report $\pi(x)$ at the quartiles of x, comparable across units. After centering x, the intercept is the logit at $\bar{x}$, and correlations among estimates with quadratic or interaction terms drop.

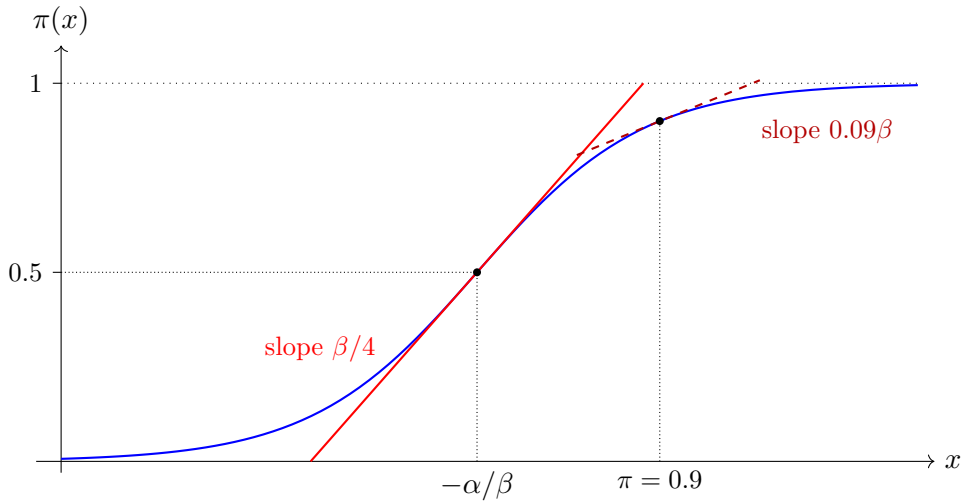


Figure 1: Logistic curve ($\alpha = -3$, $\beta = 1$) with tangent lines: slope $\beta\pi(1-\pi)$ equals $\beta/4$ at $\pi = \frac{1}{2}$ and 0.09β at $\pi = 0.9$.

5.1.2 Looking at the data

Before fitting, plot sample proportions or logits, not 0/1 responses. The sample logit $\log[y_i/(n_i - y_i)]$ is infinite when $y_i = 0$ or n_i; the adjusted version

$$\log \frac{y_i + \frac{1}{2}}{n_i - y_i + \frac{1}{2}}$$

is least biased of this form (bias $O(n_i^{-2})$, Haldane 1956). The plot should be roughly linear; with continuous x and small n_i, group nearby x or smooth and look for departures from an S-shape.

5.1.3 Horseshoe crabs revisited

Example 5.3 (Horseshoe crabs: width and presence of satellites)

Crabs (Example 4.5), $Y = 1$ if at least one satellite. The GAM fit and the proportions in the eight width categories show an increasing S-shape. With x = width (cm),

$$\hat{\pi}(x) = \frac{\exp(-12.351 + 0.497x)}{1 + \exp(-12.351 + 0.497x)}.$$

Parameter	Estimate	SE	LR 95% limits	Wald χ^2
Intercept	−12.3508	2.6287	(−17.8097, −7.4573)	22.07
width	0.4972	0.1017	(0.3084, 0.7090)	23.89

Deviance 194.45 and Pearson $X^2 = 165.14$ on 171 df; log likelihood −97.23.

(a) At the mean width 26.3, $\hat{\pi} = 0.674$; $\hat{\pi} = \frac{1}{2}$ at $x = 12.351/0.497 = 24.8$.
(b) The odds multiply by $e^{0.497} = 1.64$ per cm (a 64% increase).
(c) At $\bar{x}$ the probability rises by about $0.497(0.674)(0.326) = 0.11$ per cm.
(d) At the width quartiles 24.9, 26.1, 27.7, $\hat{\pi} = 0.51$, 0.65, 0.81: an increase of 0.30 over the middle half.

(e) With weight (kg) instead, $\operatorname{logit}\hat\pi = -3.695 + 1.815x$; $\hat\beta$'s are not comparable across units, but at the weight quartiles 2.00, 2.35, 2.85 we get $\hat\pi = 0.48,\ 0.64,\ 0.81$, an increase of 0.33, similar to width.

5.1.4 Logistic regression with retrospective studies

In case–control studies Y is fixed by design and X observed; logistic effects are odds ratios, which are estimable retrospectively.

Theorem 5.4 (Logistic regression under case–control sampling)

Let $Z = 1$ indicate that a subject is sampled, and suppose the sampling probabilities depend only on y: $P(Z = 1 \mid y, x) = \rho_y$ for $y = 0, 1$. If $\operatorname{logit} P(Y = 1 \mid x) = \alpha + \beta x$, then

$$\operatorname{logit} P(Y = 1 \mid Z = 1, x) = \alpha^* + \beta x, \qquad \alpha^* = \alpha + \log(\rho_1/\rho_0).$$

So the slope β is the same as in the prospective model; only the intercept changes.

Proof.

Step 1. By Bayes' theorem,

$$P(Y = 1 \mid Z = 1, x) = \frac{P(Z = 1 \mid Y = 1, x)P(Y = 1 \mid x)}{\sum_{j=0}^{1} P(Z = 1 \mid Y = j, x)P(Y = j \mid x)} = \frac{\rho_1 P(Y = 1 \mid x)}{\rho_0 P(Y = 0 \mid x) + \rho_1 P(Y = 1 \mid x)}.$$

Step 2. Divide numerator and denominator by $P(Y = 0 \mid x)$ and use $P(Y = 1 \mid x)/P(Y = 0 \mid x) = \exp(\alpha + \beta x)$:

$$P(Y = 1 \mid Z = 1, x) = \frac{\rho_1 \exp(\alpha + \beta x)}{\rho_0 + \rho_1 \exp(\alpha + \beta x)} = \frac{\exp[\alpha + \log(\rho_1/\rho_0) + \beta x]}{1 + \exp[\alpha + \log(\rho_1/\rho_0) + \beta x]}.$$

■

Sampling cases at 10 times the rate of controls raises the intercept by $\log 10 = 2.3$. Other binary links are not recoverable from case–control data, a main reason for the logit's popularity in biomedicine. Matched studies need conditional methods (Section 10.2.5).

Proposition 5.5 (Normal discriminant model implies logistic regression (Cornfield 1962))

Suppose $X \mid Y = i \sim N(\mu_i, \sigma^2)$, $i = 0, 1$, and $P(Y = 1) = \rho$. Then $P(Y = 1 \mid X = x)$ satisfies (5.1) with

$$\beta = \frac{\mu_1 - \mu_0}{\sigma^2}, \qquad \alpha = \log\frac{\rho}{1-\rho} - \frac{\mu_1^2 - \mu_0^2}{2\sigma^2}.$$

If the two variances differ, the logit is quadratic in x.

Proof. [Supplement] Let φ_i be the $N(\mu_i, \sigma_i^2)$ density. By Bayes' theorem,

$$\operatorname{logit} P(Y = 1 \mid x) = \log\frac{\rho\varphi_1(x)}{(1-\rho)\varphi_0(x)} = \log\frac{\rho}{1-\rho} + \log\frac{\sigma_0}{\sigma_1} - \frac{(x-\mu_1)^2}{2\sigma_1^2} + \frac{(x-\mu_0)^2}{2\sigma_0^2}.$$

With $\sigma_0 = \sigma_1 = \sigma$ the x^2 terms cancel and

$$-\frac{(x-\mu_1)^2 - (x-\mu_0)^2}{2\sigma^2} = \frac{(\mu_1 - \mu_0)x}{\sigma^2} - \frac{\mu_1^2 - \mu_0^2}{2\sigma^2},$$

giving the stated α, β. If $\sigma_0 \neq \sigma_1$ the coefficient of x^2 is $\frac{1}{2}(\sigma_0^{-2} - \sigma_1^{-2}) \neq 0$. ■

So two roughly normal groups with similar spread give a good logistic fit; unequal spreads make $\pi(x)$ nonmonotone.

5.2 Inference for logistic regression

Motivation. The Wald–LR–score triad of Theorem 4.18 applies; LR is most reliable, and goodness of fit with continuous predictors needs care.

5.2.1 Types of inference

For logit $\pi(x) = \alpha + \beta x$, the test of independence $H_0 : \beta = 0$ can use

(i) Wald: $z = \hat{\beta}/\widehat{\text{se}}$, or $z^2 \sim \chi^2_1$;

(ii) likelihood ratio: $-2(L_0 - L_1)$, twice the difference of maximized log likelihoods under H_0 and the full model, χ^2_1;

(iii) score: the derivative of the log likelihood at $\beta = 0$, comparing the sufficient statistic with its null expectation (Section 5.3.5).

They agree in large samples. LR is preferred; for large $|\beta|$ the Wald test loses power (Hauck and Donner 1977). Wald intervals are $\hat{\beta} \pm z_{\alpha/2}\widehat{\text{se}}$; inverting LR gives the profile-likelihood interval.

Proposition 5.6 (Interval for $\pi(x_0)$)

The estimated variance of the fitted logit at x_0 is

$$\widehat{\text{Var}}(\hat{\alpha} + \hat{\beta}x_0) = \widehat{\text{Var}}(\hat{\alpha}) + x_0^2\,\widehat{\text{Var}}(\hat{\beta}) + 2x_0\,\widehat{\text{Cov}}(\hat{\alpha}, \hat{\beta}),$$

and if (l, u) is a $1-\alpha$ interval for logit $\pi(x_0)$, then $\big(e^l/(1+e^l),\ e^u/(1+e^u)\big)$ is a $1-\alpha$ interval for $\pi(x_0)$.

Proof. The first formula is $\text{Var}[(1, x_0)(\hat{\alpha}, \hat{\beta})^\top] = (1, x_0)\widehat{\text{Cov}}(\hat{\alpha}, \hat{\beta})(1, x_0)^\top$. As $t \mapsto e^t/(1+e^t)$ is strictly increasing, the coverage is preserved. ■

Equivalently, refit with $x^* = x - x_0$: the intercept and its SE are the fitted logit at x_0 and its SE.

5.2.2 Inference for horseshoe crab data

Example 5.7 (Horseshoe crabs: tests and intervals)

(a) Wald: $z = 0.497/0.102 = 4.9$, $z^2 = 23.9$ ($P < 0.0001$). LR: $-2(-112.88 + 97.23) = 31.3$ (df $= 1$), stronger still.
(b) Wald interval for β: $0.497 \pm 1.96(0.102) = (0.298, 0.697)$; profile: $(0.308, 0.709)$. Odds ratio per cm: $(1.36, 2.03)$.
(c) At $x = 26.5$: logit $= -12.351 + 0.497(26.5) = 0.825$, $\hat{\pi} = 0.695$. With $\widehat{\text{Var}}(\hat{\alpha}) = 6.910$, $\widehat{\text{Var}}(\hat{\beta}) = 0.01035$, $\widehat{\text{Cov}} = -0.2668$,

$$\widehat{\text{Var}} = 6.910 + (26.5)^2(0.01035) + 2(26.5)(-0.2668) = 0.038,$$

so the logit interval is $0.825 \pm 1.96\sqrt{0.038} = (0.44, 1.21)$, i.e. $(0.61, 0.77)$ for $\pi(26.5)$.
(d) At $x = 26.5$ (6 crabs, 4 with satellites) the sample proportion 0.67 has SE $=$

$\sqrt{0.67(0.33)/6} = 0.19$ and score interval (0.30, 0.90), versus model SE = 0.04: the model borrows strength from all 173 observations (Section 6.4.5).

5.2.3 Checking goodness of fit: ungrouped and grouped data

Lack of fit is detected by an LR test against a more complex model (quadratic term, interaction), or globally by X^2, G^2 on observed versus fitted counts $n_i\hat{\pi}_i$, $n_i(1-\hat{\pi}_i)$: χ^2 with df = (settings) − (parameters) for a fixed number of settings with growing counts. This needs categorical predictors (Section 4.5.3).

5.2.4 Goodness of fit of model for horseshoe crabs

Example 5.8 (Horseshoe crabs: goodness of fit)

(a) Quadratic in centered width: $\operatorname{logit}\hat{\pi} = 0.618 + 0.533x + 0.040x^2$, $\widehat{\text{se}} = 0.046$ for x^2; LR = 0.83 (df = 1): not needed.
(b) Width has 66 distinct values; a 66×2 table violates the chi-squared theory. Using the eight width categories of Example 4.5, fitted values summed over crabs in each category:

Width (cm)	Yes	No	Fitted yes	Fitted no
< 23.25	5	9	3.64	10.36
23.25–24.25	4	10	5.31	8.69
24.25–25.25	17	11	13.78	14.22
25.25–26.25	21	18	24.23	14.77
26.25–27.25	15	7	15.94	6.06
27.25–28.25	20	4	19.38	4.62
28.25–29.25	15	3	15.65	2.35
> 29.25	14	0	13.08	0.92

$X^2 = 5.3$, $G^2 = 6.2$, df $= 8 - 2 = 6$ ($P > 0.4$): no lack of fit (approximate, as $\pi(x)$ varies within a category).

5.2.5 Checking goodness of fit with ungrouped data by grouping

With many predictors, grouping each variable gives a sparse table; group instead by fitted probabilities.

Definition (Hosmer–Lemeshow statistic)

Order observations by $\hat{\pi}$ (ungrouped fit) into g groups of about equal size (usually $g = 10$). With y_{ij}, $\hat{\pi}_{ij}$ the outcome and fitted probability of observation j in group i ($j = 1, \ldots, n_i$),

$$\text{HL} = \sum_{i=1}^{g} \frac{\left(\sum_j y_{ij} - \sum_j \hat{\pi}_{ij}\right)^2}{\left(\sum_j \hat{\pi}_{ij}\right)\left[1 - \left(\sum_j \hat{\pi}_{ij}\right)/n_i\right]}.$$

Proposition 5.9 (HL as a Pearson statistic)

HL equals the Pearson statistic $\sum(\text{observed} - \text{fitted})^2/\text{fitted}$ for the $g \times 2$ table of observed and fitted counts of successes and failures in the groups.

Proof. [Supplement] In group i let $O_i = \sum_j y_{ij}$, $E_i = \sum_j \hat{\pi}_{ij}$. The failure cell has observed $n_i - O_i$ and fitted $n_i - E_i$, so its deviation is $-(O_i - E_i)$. The two Pearson terms of row i add to

$$(O_i - E_i)^2\Big(\frac{1}{E_i} + \frac{1}{n_i - E_i}\Big) = \frac{(O_i - E_i)^2\, n_i}{E_i(n_i - E_i)} = \frac{(O_i - E_i)^2}{E_i(1 - E_i/n_i)}.$$

Summing over i gives HL. ■

Reference distribution and limitations

HL is not chi-squared in the usual sense (probabilities vary within a group); by simulation, with n distinct covariate patterns its null distribution is about χ^2_{g-2} (Hosmer–Lemeshow). Ties in $\hat{\pi}$ make software values differ slightly. Crab width model, $g = 10$: HL $= 3.5$ on 8 df, a decent fit. HL has low power and does not say what is wrong; comparisons with richer models and diagnostics (Section 6.2) are more informative.

5.3 Logit models with categorical predictors

Motivation. Codings of dummy variables change parameters, not fit or odds ratios; ordered factors with scores give the linear logit model and the Cochran–Armitage test.

5.3.1 ANOVA-type representation of factors

For a single factor X with I categories, let $y_i \sim \text{bin}(n_i, \pi_i)$ in row i of an $I \times 2$ table. The model

$$\log \frac{\pi_i}{1 - \pi_i} = \alpha + \beta_i, \qquad i = 1, \ldots, I, \tag{5.2}$$

resembles one-way ANOVA and has one redundant parameter, so a constraint is imposed, e.g. $\beta_I = 0$.

Proposition 5.10 (Coding invariance and saturation)

(a) Model (5.2) is saturated: for any $\{\pi_i\} \subset (0, 1)$ there are parameters satisfying it, and its fitted values are the sample proportions.
(b) Under any constraint, $\alpha + \beta_i = \text{logit}\,\pi_i$ and $\beta_a - \beta_b$ is the log odds ratio for rows a, b; these and the fit are the same for every coding. With $\beta_I = 0$, $\alpha = \text{logit}\,\pi_I$ and β_i is the log odds ratio of row i versus row I.
(c) No effect, $\beta_1 = \cdots = \beta_I$, is statistical independence of X and Y.

Proof.

(a) Take $\alpha = \text{logit}\,\pi_I$ and $\beta_i = \text{logit}\,\pi_i - \text{logit}\,\pi_I$. The model has I free parameters for I binomials, so it is saturated and $\hat{\pi}_i = y_i/n_i$.

(b) If $(\alpha, \boldsymbol{\beta})$ satisfies the model, so does $(\alpha - c, \beta_i + c)$ for any c, and every constraint selects one such c; $\alpha + \beta_i$ and $\beta_a - \beta_b$ do not depend on c. In particular $\beta_a - \beta_b = \text{logit}\,\pi_a - \text{logit}\,\pi_b$, a log odds ratio.

(c) Equal β_i means equal π_i across rows. ■

5.3.2 Dummy variables in logit models

Equivalently, with dummy variables $x_i = 1$ for row i and 0 otherwise ($i = 1, \ldots, I-1$),

$$\operatorname{logit} \pi = \alpha + \beta_1 x_1 + \cdots + \beta_{I-1} x_{I-1};$$

no dummy for category I means $\beta_I = 0$ (some software sets $\beta_1 = 0$). *Effect coding* imposes $\sum_i \beta_i = 0$; for $I = 2$, $x = \pm 1$ and the log odds ratio is $2\beta_1$ (versus β_1 for a 0/1 dummy). A single $\hat{\beta}_i$ is meaningful only relative to another category.

5.3.3 Alcohol and infant malformation example revisited

Example 5.11 (Maternal alcohol consumption and malformation)

Alcohol (drinks/day)	Present	Absent	Logit	Observed prop.	Fitted (linear logit)
0	48	17066	−5.87	0.0028	0.0026
< 1	38	14464	−5.94	0.0026	0.0030
1–2	5	788	−5.06	0.0063	0.0041
3–5	1	126	−4.84	0.0079	0.0091
≥ 6	1	37	−3.61	0.0263	0.0231

For (5.2), $\hat{\alpha} + \hat{\beta}_i$ are the sample logits, e.g. $\log(48/17066) = -5.87$. With $\beta_5 = 0$: $\hat{\alpha} = -3.61$, $\hat{\beta}_1 = -2.26$; with $\beta_1 = 0$: $\hat{\alpha} = -5.87$. Apart from the first two rows, logits increase with alcohol. Independence ($\hat{\alpha} = \log(93/32481) = -5.86$, df = 4): $X^2 = 12.1$ ($P = 0.02$), $G^2 = 6.2$ ($P = 0.19$), mixed because the counts are very unbalanced (exact P: 0.03, 0.13). Both ignore the ordering, and one case more or less in the last rows changes those proportions a lot.

5.3.4 Linear logit model for $I \times 2$ tables

For ordered categories with scores $x_1, \ldots, x_I$ the *linear logit model*

$$\operatorname{logit} \pi_i = \alpha + \beta x_i \tag{5.3}$$

lies between independence ($\beta = 0$) and the saturated model (5.2).

Example 5.12 (Alcohol: linear logit fit)

Scores (0, 0.5, 1.5, 4.0, 7.0): $\hat{\alpha} = -5.9605$ (0.1154), $\hat{\beta} = 0.3166$ (0.1254), profile interval (0.0187, 0.5236); odds multiply by $e^{0.317} = 1.37$ per drink per day. $G^2 = 1.95$, $X^2 = 2.05$, df = 3; log likelihood −635.60. Fitted proportions: Example 5.11.

5.3.5 Cochran–Armitage trend test

For an $I \times 2$ table with ordered rows and independent $y_i \sim \text{bin}(n_i, \pi_i)$, Cochran (1954) and Armitage (1955) partitioned X^2 via $\pi_i = \alpha + \beta x_i$ fitted by weighted least squares. With $n = \sum n_i$, $p_i = y_i/n_i$, $p = \sum y_i/n$, $\bar{x} = \sum n_i x_i/n$, the fit is $\hat{\pi}_i = p + b(x_i - \bar{x})$ with

$$b = \frac{\sum_i n_i (p_i - p)(x_i - \bar{x})}{\sum_i n_i (x_i - \bar{x})^2}.$$

Theorem 5.13 (Partition of the Pearson statistic)

Let $X^2(I) = \sum_i n_i(p_i - p)^2/[p(1-p)]$ be the Pearson statistic for independence in the $I \times 2$ table. Then

$$X^2(I) = z^2 + X^2(L),$$

where

$$X^2(L) = \frac{\sum_i n_i(p_i - \hat{\pi}_i)^2}{p(1-p)}, \qquad z^2 = \frac{b^2 \sum_i n_i(x_i - \bar{x})^2}{p(1-p)} = \frac{\left[\sum_i (x_i - \bar{x})y_i\right]^2}{p(1-p)\sum_i n_i(x_i - \bar{x})^2}.$$

Under independence z^2 is asymptotically χ^2_1 (the *Cochran–Armitage trend test* of $\beta = 0$); when the linear probability model holds, $X^2(L)$ is asymptotically χ^2_{I-2} (a test of linearity).

Proof. [Supplement]

Step 1. (The Pearson form.) The cells of row i have observed y_i, $n_i - y_i$ and fitted $n_i p$, $n_i(1-p)$, with deviations $\pm(y_i - n_i p)$. As in Proposition 5.9 the two terms add to $(y_i - n_i p)^2/[n_i p(1-p)] = n_i(p_i - p)^2/[p(1-p)]$.

Step 2. (Decomposition.) Write $p_i - p = (p_i - \hat{\pi}_i) + (\hat{\pi}_i - p)$. Then

$$\sum n_i(p_i - p)^2 = \sum n_i(p_i - \hat{\pi}_i)^2 + \sum n_i(\hat{\pi}_i - p)^2 + 2\sum n_i(p_i - \hat{\pi}_i)(\hat{\pi}_i - p).$$

Step 3. (Cross term vanishes.) Since $\hat{\pi}_i - p = b(x_i - \bar{x})$,

$$\sum n_i(p_i - \hat{\pi}_i)(x_i - \bar{x}) = \sum n_i(p_i - p)(x_i - \bar{x}) - b\sum n_i(x_i - \bar{x})^2 = 0$$

by the definition of b.

Step 4. (Middle term.) $\sum n_i(\hat{\pi}_i - p)^2 = b^2 \sum n_i(x_i - \bar{x})^2$. Since $\sum n_i(x_i - \bar{x}) = 0$, $\sum n_i(p_i - p)(x_i - \bar{x}) = \sum (y_i - n_i p)(x_i - \bar{x}) = \sum (x_i - \bar{x})y_i$, which gives the second form of z^2. Dividing Step 2 by $p(1-p)$ yields the partition.

Step 5. (Null distribution of z^2.) Under independence $E\sum(x_i - \bar{x})Y_i = \pi \sum n_i(x_i - \bar{x}) = 0$ and $\operatorname{Var}\sum(x_i - \bar{x})Y_i = \pi(1-\pi)\sum n_i(x_i - \bar{x})^2$. By the CLT and $p \xrightarrow{p} \pi$ (Slutsky), $z \xrightarrow{d} N(0,1)$. The limit of $X^2(L)$ follows from the chi-squared theory for the fit of a 2-parameter model to I binomials. ■

Proposition 5.14 (Cochran–Armitage = score test)

The score statistic for $H_0 : \beta = 0$ in the linear logit model (5.3) equals z^2 of Theorem 5.13.

Proof. [Supplement]

Step 1. Under H_0 the MLE is $\tilde{\alpha} = \operatorname{logit} p$, so $\tilde{\pi}_i = p$. By (4.16) in count form the score components at $(\tilde{\alpha}, 0)$ are

$$u_\alpha = \sum_i (y_i - n_i p) = 0, \qquad u_\beta = \sum_i x_i(y_i - n_i p) = \sum_i (x_i - \bar{x})y_i.$$

Step 2. The information at $(\tilde{\alpha}, 0)$ is $X^\top \operatorname{diag}\{n_i p(1-p)\} X$:

$$\mathcal{I} = p(1-p)\begin{pmatrix} n & \sum n_i x_i \\ \sum n_i x_i & \sum n_i x_i^2 \end{pmatrix},$$

and by the formula for the inverse of a 2×2 matrix,

$$[\mathcal{I}^{-1}]_{\beta\beta} = \frac{n}{p(1-p)\left[n\sum n_i x_i^2 - (\sum n_i x_i)^2\right]} = \frac{1}{p(1-p)\sum n_i(x_i - \bar{x})^2}.$$

Step 3. The score statistic $\boldsymbol{u}^\top \mathcal{I}^{-1}\boldsymbol{u}$ with $u_\alpha = 0$ is $u_\beta^2[\mathcal{I}^{-1}]_{\beta\beta} = z^2$. ■

The statistic also equals $M^2 = (n-1)r^2$ of Section 3.4 for $J = 2$, with $n-1$ replaced by n; for $I = 2$, $X^2(L) = 0$ and $z^2 = X^2(I)$.

Example 5.15 (Alcohol: trend test)

Here $z^2 = 6.6$ ($P = 0.010$) and $X^2(I) = 12.1 = 6.6 + 5.5$, $X^2(L) = 5.5$ (df $= 3$): slight nonlinearity. $r = 0.014$ ($n = 32{,}574$) is a poor descriptive measure for such unbalanced tables. Wald $(0.317/0.125)^2 = 6.4$ ($P = 0.012$), LR 4.25 ($P = 0.039$); LR is safest here, and the profile interval (0.02, 0.52) is preferable to Wald (0.07, 0.56). Exact methods (Section 6.7.4) matter even for huge n.

5.4 Multiple logistic regression

Motivation. Each coefficient is a conditional effect; deviance differences of nested models decide which terms are needed.

For predictors $\mathbf{x} = (x_1, \ldots, x_p)$,

$$\operatorname{logit}\pi(\mathbf{x}) = \alpha + \beta_1 x_1 + \cdots + \beta_p x_p, \qquad \pi(\mathbf{x}) = \frac{\exp(\alpha + \beta_1 x_1 + \cdots + \beta_p x_p)}{1 + \exp(\alpha + \beta_1 x_1 + \cdots + \beta_p x_p)}. \tag{5.4}$$

e^{β_j} is the multiplicative effect on the odds of a unit increase in x_j, at fixed levels of the other predictors.

5.4.1 Logit models for multiway contingency tables

With binary X and Z coded by dummies ($x_1 = z_1 = 1$, $x_2 = z_2 = 0$) and independent binomials at each (i, k), consider

$$\operatorname{logit} P(Y = 1) = \alpha + \beta_1 x_i + \beta_2 z_k. \tag{5.5}$$

Proposition 5.16 (Main-effects model and homogeneous association)

In (5.5), at each level z_k of Z,

$$\operatorname{logit} P(Y = 1 \mid x = 1, z_k) - \operatorname{logit} P(Y = 1 \mid x = 0, z_k) = \beta_1,$$

so e^{β_1} is the conditional XY odds ratio, the same at each level of Z (homogeneous association). $\beta_1 = 0$ is conditional independence of X and Y given Z.

Proof. $(\alpha + \beta_1 + \beta_2 z_k) - (\alpha + \beta_2 z_k) = \beta_1$, free of z_k; a difference of logits is a log odds ratio. If $\beta_1 = 0$ the conditional odds ratios all equal 1. ■

No interaction means additivity on the logit scale; on another scale interaction may appear or disappear. For factors with any numbers of categories:

$$\operatorname{logit} P(Y = 1) = \alpha + \beta_i^X + \beta_k^Z, \tag{5.6}$$

with one redundant parameter per factor (e.g. $\beta_I^X = \beta_K^Z = 0$; superscripts are labels). Conditional independence of X and Y is $\beta_1^X = \cdots = \beta_I^X$. For two categories each, (5.6) with $\beta_2^X = \beta_2^Z = 0$ is (5.5) with $\beta_1^X = \beta_1$, $\beta_1^Z = \beta_2$.

5.4.2 AIDS and AZT example

Example 5.17 (AIDS symptoms, AZT use and race)

338 veterans with faltering immune systems, randomized to immediate or deferred AZT; response: AIDS symptoms within 3 years.

		Symptoms	
Race	AZT use	Yes	No
White	Yes	14	93
	No	32	81
Black	Yes	11	52
	No	12	43

In (5.5) let $x = 1$ for immediate AZT and $z = 1$ for whites. Fit:

Parameter	Estimate	SE	Wald χ^2
Intercept	-1.0736	0.2629	16.67
AZT	-0.7195	0.2790	6.65
race	0.0555	0.2886	0.04

(a) Conditional odds ratio for AZT $e^{-0.7195} = 0.487$: odds of symptoms about halved, for each race. Wald $\exp[-0.720 \pm 1.96(0.279)] = (0.28, 0.84)$; profile (0.279, 0.835). Race: 1.057, (0.60, 1.86).
(b) $H_0 : \beta_1 = 0$ (conditional independence of AZT and symptoms given race): LR statistic 6.9 (df $= 1$, $P = 0.01$); Wald $(-0.720/0.279)^2 = 6.65$.
(c) Codings (last $= 0$, first $= 0$, sum $= 0$): intercept $(-1.074, -1.738, -1.406)$, AZT-yes $(-0.720, 0, -0.360)$, white $(0.055, 0, 0.028)$. Always intercept + AZT-yes + white $= -1.738$ ($\hat{\pi} = 0.15$, whites with immediate AZT) and $\hat{\beta}_1^X - \hat{\beta}_2^X = -0.720$.
(d) Fitted probabilities (95% limits): white/AZT 0.150 (0.099, 0.220); white/no 0.265 (0.197, 0.348); black/AZT 0.143 (0.087, 0.225); black/no 0.255 (0.170, 0.364).

5.4.3 Goodness of fit as a likelihood-ratio test

Let $G^2(M_0 \mid M_1) = -2(L_0 - L_1)$. The fit statistic $G^2(M)$ is the case M_1 = saturated, df = (binomials) − (parameters). By Proposition 4.22, with L_S the saturated log likelihood,

$$G^2(M_0 \mid M_1) = -2(L_0 - L_S) - [-2(L_1 - L_S)] = G^2(M_0) - G^2(M_1).$$

Example 5.18 (AIDS: goodness of fit)

Whites with immediate AZT: $\hat{\pi} = 0.150$, fitted counts $107(0.150) = 16.0$ and 91.0; likewise for all 8 cells. $G^2 = 1.38$, $X^2 = 1.39$, df $= 4 - 3 = 1$ ($P > 0.2$), also a test of homogeneous odds ratios. No race effect: $G^2(M_0) - G^2(M_1) = 1.42 - 1.38 = 0.04$.

The difference is often nearly chi-squared even when the separate $G^2(M_i)$ are not, if the models differ by few parameters.

5.4.4 Horseshoe crab example revisited

Example 5.19 (Horseshoe crabs: width and color)

Color (medium light, medium, medium dark, dark; a surrogate for age), dummies c_1, c_2, c_3 with dark as reference:

$$\operatorname{logit}\pi = \alpha + \beta_1 c_1 + \beta_2 c_2 + \beta_3 c_3 + \beta_4 x. \tag{5.7}$$

Parameter	Estimate	SE	LR 95% limits	χ^2
intercept	−12.7151	2.7618	(−18.4564, −7.5788)	21.20
c_1	1.3299	0.8525	(−0.2738, 3.1354)	2.43
c_2	1.4023	0.5484	(0.3527, 2.5260)	6.54
c_3	1.1061	0.5921	(−0.0279, 2.3138)	3.49
width	0.4680	0.1055	(0.2713, 0.6870)	19.66

Deviance 187.46, Pearson 168.66 on 168 df; log likelihood −93.73.

Dark: $\operatorname{logit}\hat{\pi} = -12.715 + 0.468x$; medium light: $-11.385 + 0.468x$. At $x = 26.3$, $\hat{\pi} = 0.399$ and 0.715. Without interaction the curves are horizontal shifts and never cross; 1 cm multiplies the odds by $e^{0.468} = 1.60$ for every color, and medium light vs. dark by $e^{1.330} = 3.8$ at any width.

5.4.5 Model comparison

Example 5.20 (Horseshoe crabs: testing color and interaction)

No color effect given width: LR = 7.0, df = 3, $P = 0.07$. Adding color×width (separate fits per color): LR = 4.4, df = 3, $P = 0.22$, weak evidence of interaction.

5.4.6 Quantitative treatment of ordinal predictor

Example 5.21 (Horseshoe crabs: color scores)

With scores $c = 1, 2, 3, 4$,

$$\operatorname{logit}\pi = \alpha + \beta_1 c + \beta_2 x, \qquad \hat{\beta}_1 = -0.509\ (0.224), \quad \hat{\beta}_2 = 0.458\ (0.104).$$

Each step darker multiplies the odds by $e^{-0.509} = 0.60$ at fixed width; versus (5.7), LR = 1.7 (df = 2, $P = 0.44$). The estimates (1.33, 1.40, 1.11, 0) suggest dark (0) versus other (1); versus (5.7), LR = 0.5 (df = 2), with fit

$$\operatorname{logit}\hat{\pi} = -12.980 + 1.300c + 0.478x \qquad (\widehat{\text{se}}:\ 0.526,\ 0.104). \tag{5.8}$$

Non-dark crabs have $e^{1.300} = 3.7$ times the odds of dark crabs at a given width. All three models make dark crabs least likely to have satellites. A quantitative treatment of an ordinal predictor, when it fits, is simpler and gives more powerful one-df tests (Section 6.4).

5.4.7 Standardized and probability-based interpretations

To compare predictors with different units, multiply $\hat{\beta}_j$ by s_{x_j} (a one-SD effect). On the probability scale a unit increase in x_j changes π by about $\hat{\beta}_j\hat{\pi}(1-\hat{\pi})$ (Proposition 5.2); or report $\hat{\pi}$ at the quartiles of x_j with others at their means (at both values for a dummy).

Example 5.22 (Horseshoe crabs: effects on the probability scale)

For (5.8): at $\hat{\pi} = 0.5$, 1 cm raises π by about $0.478(0.5)(0.5) = 0.12$ (1 cm $< \frac{1}{2}$ SD). With $\bar{c} = 0.873$, $\hat{\pi} = 0.51$ and 0.80 at the width quartiles 24.9, 27.7. At $x = 26.3$: 0.40 (dark), 0.71 (other).

Variable	Estimate	SE	Comparison	Change in probability
No interaction model				
Intercept	−12.980	2.727		
Color (0 dark, 1 other)	1.300	0.526	$(1,0)$ at $\bar{x}$	$0.31 = 0.71 - 0.40$
Width x (cm)	0.478	0.104	(UQ, LQ) at $\bar{c}$	$0.29 = 0.80 - 0.51$
Interaction model				
Intercept	−5.854	6.694		
Color	−6.958	7.318		
Width x	0.200	0.262	(UQ, LQ) at $c = 0$	$0.13 = 0.43 - 0.30$
Width × color	0.322	0.286	(UQ, LQ) at $c = 1$	$0.29 = 0.84 - 0.55$

The width effect appears larger for lighter crabs, but the interaction is not significant.

5.5 Fitting logistic regression models

Motivation. As a canonical-link GLM, logistic regression has a concave likelihood, Newton–Raphson = Fisher scoring = IRLS, and finite estimates exactly under overlap.

Let $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})$ be setting i ($i = 1, \ldots, N$), $x_{i1} = 1$, with y_i successes in n_i trials (continuous predictors: typically $N = n$, $n_i = 1$). The model is

$$\pi(\mathbf{x}_i) = \frac{\exp\left(\sum_{j=1}^{p} \beta_j x_{ij}\right)}{1 + \exp\left(\sum_{j=1}^{p} \beta_j x_{ij}\right)}. \tag{5.9}$$

5.5.1 Likelihood equations

Theorem 5.23 (Logistic log likelihood and likelihood equations)

For independent $y_i \sim \text{bin}(n_i, \pi(\mathbf{x}_i))$ under (5.9),

$$L(\boldsymbol{\beta}) = \sum_j \Big(\sum_i y_i x_{ij}\Big)\beta_j - \sum_i n_i \log\Big[1 + \exp\Big(\sum_j \beta_j x_{ij}\Big)\Big], \tag{5.10}$$

so $\{\sum_i y_i x_{ij}\}$ are sufficient, and the likelihood equations are

$$\sum_i y_i x_{ij} - \sum_i n_i \hat{\pi}_i x_{ij} = 0, \quad j = 1, \ldots, p, \qquad \text{i.e.} \qquad X^\top \mathbf{y} = X^\top \hat{\boldsymbol{\mu}}, \tag{5.11}$$

where $\hat{\mu}_i = n_i \hat{\pi}_i$.

Proof.

Step 1. The joint mass function is proportional to

$$\prod_i \pi_i^{y_i}(1-\pi_i)^{n_i-y_i} = \Big\{\prod_i (1-\pi_i)^{n_i}\Big\} \exp\Big[\sum_i y_i \log\frac{\pi_i}{1-\pi_i}\Big].$$

Step 2. With $\operatorname{logit}\pi_i = \sum_j \beta_j x_{ij}$, the exponent is $\sum_i y_i \sum_j \beta_j x_{ij} = \sum_j \beta_j \sum_i y_i x_{ij}$, and $1-\pi_i = [1+\exp(\sum_j \beta_j x_{ij})]^{-1}$. Taking logs gives (5.10); the factorization theorem gives sufficiency.

Step 3. Differentiating,

$$\frac{\partial L}{\partial \beta_j} = \sum_i y_i x_{ij} - \sum_i n_i x_{ij} \frac{\exp(\sum_k \beta_k x_{ik})}{1+\exp(\sum_k \beta_k x_{ik})} = \sum_i (y_i - n_i\pi_i) x_{ij}.$$

■

These are (4.16) with counts: by Theorem 4.27 they equate sufficient statistics to their expectations. They are nonlinear and need iteration.

5.5.2 Asymptotic covariance matrix of parameter estimators

Theorem 5.24 (Information matrix for logistic regression)

$$-\frac{\partial^2 L}{\partial\beta_a \partial\beta_b} = \sum_i x_{ia}x_{ib}\, n_i\pi_i(1-\pi_i),$$

which does not involve $\mathbf{y}$; so observed and expected information coincide, and

$$\widehat{\operatorname{Cov}}(\hat{\boldsymbol{\beta}}) = \{X^\top \operatorname{diag}[n_i\hat\pi_i(1-\hat\pi_i)]X\}^{-1}. \tag{5.12}$$

Proof. From Step 3 above, $\partial L/\partial\beta_a = \sum_i (y_i - n_i\pi_i)x_{ia}$, and by Proposition 5.2, Step 1, $\partial\pi_i/\partial\beta_b = \pi_i(1-\pi_i)x_{ib}$. Hence $\partial^2 L/\partial\beta_a\partial\beta_b = -\sum_i n_i\pi_i(1-\pi_i)x_{ia}x_{ib}$. This is free of y_i, so it equals its expectation, and Theorem 4.15 gives (5.12): the GLM form $(X^\top \hat W X)^{-1}$ with $w_i = n_i\pi_i(1-\pi_i)$. ■

The square roots of the diagonal of (5.12) are the standard errors of $\hat{\boldsymbol{\beta}}$.

5.5.3 Distribution of probability estimators

For a setting $\mathbf{x}$, the fitted logit $\mathbf{x}^\top\hat{\boldsymbol{\beta}}$ has estimated variance $\mathbf{x}^\top\widehat{\operatorname{Cov}}(\hat{\boldsymbol{\beta}})\mathbf{x}$, so

$$\mathbf{x}^\top\hat{\boldsymbol{\beta}} \pm z_{\alpha/2}\sqrt{\mathbf{x}^\top\widehat{\operatorname{Cov}}(\hat{\boldsymbol{\beta}})\mathbf{x}}$$

is a large-sample interval for $\operatorname{logit}\pi(\mathbf{x})$; applying $t\mapsto e^t/(1+e^t)$ to the endpoints gives one for $\pi(\mathbf{x})$ (Proposition 5.6).

5.5.4 Newton–Raphson method applied to logistic regression

Proposition 5.25 (Newton–Raphson / IRLS for logistic regression)

With $\pi_i^{(t)} = \exp(\mathbf{x}_i^\top\boldsymbol{\beta}^{(t)})/[1+\exp(\mathbf{x}_i^\top\boldsymbol{\beta}^{(t)})]$ and $\mu_i^{(t)} = n_i\pi_i^{(t)}$, the Newton–Raphson step is

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + \{X^\top \operatorname{diag}[n_i\pi_i^{(t)}(1-\pi_i^{(t)})]X\}^{-1}X^\top(\mathbf{y}-\boldsymbol{\mu}^{(t)}). \tag{5.13}$$

Equivalently $\boldsymbol{\beta}^{(t+1)} = (X^\top V_t^{-1} X)^{-1} X^\top V_t^{-1} \boldsymbol{z}^{(t)}$, with $V_t = \text{diag}\{1/[n_i \pi_i^{(t)}(1-\pi_i^{(t)})]\}$ and

$$z_i^{(t)} = \log \frac{\pi_i^{(t)}}{1-\pi_i^{(t)}} + \frac{y_i - n_i \pi_i^{(t)}}{n_i \pi_i^{(t)}(1-\pi_i^{(t)})}. \tag{5.14}$$

At convergence, $-H^{-1}$ is the estimated covariance (5.12).

Proof. By Theorems 5.23 and 5.24, $\boldsymbol{u}^{(t)} = X^\top(\mathbf{y} - \boldsymbol{\mu}^{(t)})$ and $H^{(t)} = -X^\top \text{diag}[n_i \pi_i^{(t)}(1-\pi_i^{(t)})] X$; substitute in (4.25). Since $H = -\mathcal{I}$ (canonical link), this is also Fisher scoring, hence IRLS by Theorem 4.26. For the logit, $\partial \eta_i / \partial \mu_i = 1/[n_i \pi_i (1-\pi_i)]$ when $\mu_i = n_i \pi_i$, so the working response (4.28) is (5.14), and $W = V_t^{-1}$. ■

V_t holds the estimated variances of sample logits (Section 3.1.6) and $\boldsymbol{z}^{(t)}$ is the linearized sample logit. From $\boldsymbol{\beta}^{(0)}$ (e.g. $\mathbf{0}$) the iterates converge to $\hat{\boldsymbol{\beta}}$ (Walker and Duncan 1967).

5.5.5 Convergence and existence of finite estimates

Theorem 5.26 (Concavity and uniqueness)

If X has full column rank, $L(\boldsymbol{\beta})$ in (5.10) is strictly concave; hence the ML estimate, when it exists, is unique, and any root of (5.11) is it.

Proof. By Theorem 5.24, $\boldsymbol{c}^\top H \boldsymbol{c} = -\sum_i n_i \pi_i (1-\pi_i)(\mathbf{x}_i^\top \boldsymbol{c})^2$ with every $n_i \pi_i (1-\pi_i) > 0$. It is < 0 unless $X\boldsymbol{c} = \mathbf{0}$, i.e. $\boldsymbol{c} = \mathbf{0}$. A strictly concave function has at most one maximizer, and a stationary point of a concave function is a global maximum. ■

Theorem 5.27 (Existence of the ML estimate: overlap versus separation)

Write the data as n individual binary observations $(\mathbf{x}_i, y_i)$ and let $s_i = 2y_i - 1 \in \{-1, 1\}$. Assume X has full column rank. Call the data *separated* if some $\boldsymbol{d} \neq \mathbf{0}$ satisfies $s_i \mathbf{x}_i^\top \boldsymbol{d} \geq 0$ for all i, with strict inequality for at least one i (a hyperplane with all $y = 1$ on one side and all $y = 0$ on the other, ties on it allowed). Otherwise the data *overlap*.
(a) If the data are separated, no ML estimate exists: L increases indefinitely along $\boldsymbol{d}$ and some $|\hat{\beta}_j|$ is "infinite".
(b) If the data overlap, a finite ML estimate exists, and it is unique.

Proof. [Supplement]

Step 1. (A convenient form.) For one binary observation, $y\eta - \log(1+e^\eta) = -\log(1+e^{-s\eta})$ (check $y = 1$ and $y = 0$). Hence

$$L(\boldsymbol{\beta}) = -\sum_i \log[1 + \exp(-s_i \mathbf{x}_i^\top \boldsymbol{\beta})] < 0.$$

Each term is a strictly increasing function of $s_i \mathbf{x}_i^\top \boldsymbol{\beta}$.

Step 2. (a) Suppose $\boldsymbol{d}$ separates and $\hat{\boldsymbol{\beta}}$ were a maximizer. For $t > 0$, $s_i \mathbf{x}_i^\top (\hat{\boldsymbol{\beta}} + t\boldsymbol{d}) \geq s_i \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}$ for all i, with strict inequality for some i. By Step 1, $L(\hat{\boldsymbol{\beta}} + t\boldsymbol{d}) > L(\hat{\boldsymbol{\beta}})$, a contradiction. So the supremum is not attained; along $\boldsymbol{\beta} + t\boldsymbol{d}$ the likelihood keeps increasing as $t \to \infty$ (under complete separation, $s_i \mathbf{x}_i^\top \boldsymbol{d} > 0$ for all i, it tends to the perfect-fit value $L = 0$).

Step 3. (b) Overlap and full rank imply: for every unit vector $\boldsymbol{d}$, $m(\boldsymbol{d}) = \min_i s_i \mathbf{x}_i^\top \boldsymbol{d} < 0$. Indeed, otherwise $s_i \mathbf{x}_i^\top \boldsymbol{d} \geq 0$ for all i; not separated forces all equalities, so $X\boldsymbol{d} = \mathbf{0}$, contradicting full rank.

Step 4. m is continuous on the compact unit sphere, so $\max_{\|\boldsymbol{d}\|=1} m(\boldsymbol{d}) = -c < 0$. For $\boldsymbol{\beta} = r\boldsymbol{d}$

with $r > 0$, $\|\boldsymbol{d}\| = 1$, choose i with $s_i \mathbf{x}_i^\top \boldsymbol{d} \le -c$; then by Step 1

$$L(r\boldsymbol{d}) \le -\log[1 + \exp(rc)] \le -rc.$$

Step 5. Hence $\{\boldsymbol{\beta} : L(\boldsymbol{\beta}) \ge L(\mathbf{0})\} \subset \{\|\boldsymbol{\beta}\| \le -L(\mathbf{0})/c\}$ is bounded; it is closed by continuity, hence compact and nonempty. The continuous L attains its maximum on it, which is a global maximum. Uniqueness is Theorem 5.26. ■

Similar results hold for the probit and some other links (Silvapulle 1981). Software rarely detects $\hat{\beta} = \pm\infty$: the log likelihood flattens, convergence criteria are met, and enormous standard errors are reported.

Example 5.28 (Perfect discrimination)

$y = 0$ at $x = 10, 20, 30, 40$, $y = 1$ at $x = 60, 70, 80, 90$. With $\alpha = -50\beta$ ($\hat{\pi}(50) = 0.5$) and $\beta \to \infty$ the likelihood increases toward a perfect fit (Figure 2). PROC GENMOD reports $\operatorname{logit} \hat{\pi} = -192.2 + 3.8x$ with standard errors 8.0×10^8 and 1.5×10^7, artifacts of $\hat{\beta} = \infty$.

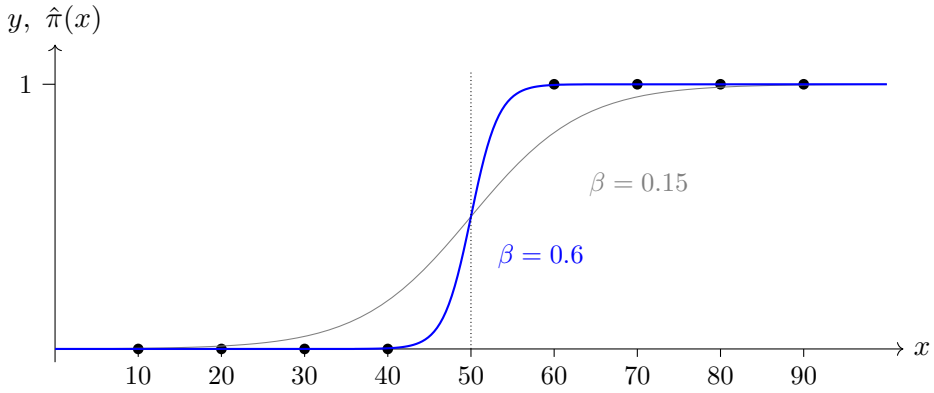


Figure 2: Complete separation: with $\alpha = -50\beta$ the curve sharpens to a step at $x = 50$ as $\beta \to \infty$; the likelihood has no maximum.